

Code Layout, Readability and Reusability

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist

| S.No | Code Quality Parameter    | Description                         | Followed | Remarks                        |
|------|---------------------------|-------------------------------------|----------|--------------------------------|
| 1    | Consistent Indentation    | Uniform indentation/formatting      | Yes      | Properly formatted Python code |
| 2    | Proper File Structure     | Files/folders logically organized   | Yes      | Easy to navigate               |
| 3    | Meaningful Variable Names | Variables reflect purpose clearly   | Yes      | Self-explanatory               |
| 4    | Function/Method Names     | Functions are descriptively named   | Yes      | Improves understanding         |
| 5    | Code Comments             | Inline/block comments explain logic | Yes      | Enhances maintainability       |
| 6    | Modular Design            | Code split into reusable modules    | Yes      | Separate logic modules         |
| 7    | No Redundant Code         | Duplicate/unused code removed       | Yes      | Efficient implementation       |
| 8    | Error Handling            | Exceptions handled gracefully       | Yes      | Prevents crashes               |

Reusable Components / Modules

| S.No | Component / Module Name  | Language / Technology | Where Used            | Reusability Level |
|------|--------------------------|-----------------------|-----------------------|-------------------|
| 1    | app.py                   | Python, Streamlit     | Main app workflow     | High              |
| 2    | BiLSTM Model             | TensorFlow, Keras     | Emotion prediction    | High              |
| 3    | Tokenizer                | Keras                 | Text preprocessing    | High              |
| 4    | Label Encoder            | Scikit-learn          | Emotion decoding      | High              |
| 5    | Learning Guidance Module | Python, Gemini API    | Personalized guidance | Medium            |

Overall Code Quality Assessment

| Aspect                   | Rating (1-5) | Comments                   |
|--------------------------|--------------|----------------------------|
| Code Layout & Structure  | 5            | Well-organized directory   |
| Readability              | 5            | Clean, consistent style    |
| Reusability              | 5            | Easily reusable components |
| Documentation / Comments | 4            | Sufficient comments        |
| Overall Score            | 5            | Modular and maintainable   |